DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

HERIOT-WATT UNIVERSITY

F78AB Actuarial and Financial Mathematics B

Date: 11 May 2015

Time: 09:00 to 11:00

Question	Marks
1	7
2	6
3	11
4	12
5	14
6	10
Total Marks	60

Attempt ALL questions.

Show all steps in your calculation.

Approved electronic calculators may be used.

1. The annual force of interest is given by $\delta(t) = 0.005t$, for all $t \in [0, 6]$, and time is measured in years.

(a) Calculate the present value at time 0 of £100 at time $t = 5$. [2 marks]

(b) Calculate the accumulated value at time $t = 6$ of a continuous payment stream, which is paid from time $t = 3$ until time $t = 6$ at the rate £$10t$ per annum. [5 marks]

[Total 7 Marks]

2. (a) Explain what is meant by short-selling a financial asset. [1 mark]

(b) A 7-month forward contract is issued on 1 May 2015 on a stock with a price of £60. Dividends of £2 per unit holding of the stock are expected after 3 and 6 months.

The interest rate on both deposits and loans is 4% per annum, expressed as a force of interest.

By using a replicating portfolio argument, calculate the forward price.

State any assumption that you make.

[5 marks]

[Total 6 Marks]

3. At time $t = 0$ an investor purchased an annuity-certain which paid her £20 000 per annum annually in arrears for three years. The purchase price paid by the investor was £54 000.

The value of the retail price index at various times was as shown in the table below.

Time t (years)	0	1	2	3
Retail price index	341.4	366.6	382.0	401.8

(a) Calculate the following effective rates of return per annum achieved by the investor from her investment in the annuity, rounded to the nearest 0.5%:

(i) the real rate of return.

(ii) the money rate of return.

[7 marks]

(b) A colleague says that the average rate of inflation over the three-year period can be calculated from the real and money rates of return only.

State with reasons whether you agree or disagree with your colleague.

[4 marks]

[Total 11 Marks]

[PLEASE TURN OVER]

4. Denote the rate of return on an investment fund in year n, i.e. over the time interval $[n-1, n)$, by i_n. It is assumed that $i_1, i_2, \ldots$ are independent and identically distributed random variables such that $1 + i_1$ has a lognormal distribution with parameters μ and σ^2.

Based on historical data, the parameters are chosen so that the mean of i_1 equals 0.07 and the standard deviation of i_1 equals 0.15.

If a random variable X is lognormally distributed with parameters μ and σ^2 then the kth moment of X is $e^{k\mu+\frac{1}{2}k^2\sigma^2}$, for $k = 1, 2, 3, \ldots$.

(a) Calculate the values of μ and σ.

[3 marks]

At the start of year 1, an investor invests £5 000 in the investment fund for 20 years. Denote by S_{20} the accumulation of £1 from the start of year 1 to the end of year 20.

(b) Determine the distribution of $5\,000S_{20}$, including its parameters.

[3 marks]

(c) Calculate:

(i) the mean accumulated value at the end of year 20.

(ii) the corresponding mean 20-year return on the investment, expressed as an annual force of interest.

(iii) the median accumulated value at the end of year 20.

(iv) the corresponding median 20-year return on the investment, expressed as an annual force of interest.

Hint: the cumulative distribution function Φ of the standard normal distribution satisfies $\Phi(0) = 0.5$.

[4 marks]

(d) The investor isn't familiar with probability distributions and asks you what their annualised investment return will be at the end of 20 years.

State in no more than 50 words what you would tell the investor. Refer to either the median or the mean return in your response, but not both.

[2 marks]

[Total 12 Marks]

[PLEASE TURN OVER]

5. (a) An investment provides income of £1 million payable at the end of each year for the next 12 years. There is no capital repayment.

If the interest rate is 5% per annum effective, show that the duration of the investment is 5.922 years.

[5 marks]

(b) An investment company has liabilities of £10 million due in 7 years' time and £11 million due in 9 years' time.

The company holds two investments, A and B. Investment A is the investment described in part (a) and Investment B is a zero-coupon bond which pays £X at the end of N years (where N is not necessarily an integer).

The interest rate is 5% per annum effective. You are given that $\sum_{k=1}^{12} k^2 v^k = 414.647$ at 5%.

Investigate whether values of £X and N can be found which ensure that the investment company is immunized against small changes in the interest rate.

[9 marks]

[Total 14 Marks]

6. Expressed as an annual effective rate, the n-year spot rate of interest is

$$y_n = 0.006 + \frac{6n}{1000},$$

for $n = 1, 2, \ldots$.

(a) Calculate the implied 5-year forward rate from time $n = 2$, expressed as an annual effective rate. State any assumption used.

[4 marks]

A project provides the cashflows in the table below.

Time n	0	1	2	3	4
Cashflow C_n (£)	−65 000	5 000	10 000	30 000	30 000

Using the spot rates $y_1, y_2, \ldots$ given above,

(b) Calculate the net present value of the project.

[4 marks]

(c) Determine the discounted payback period of the project.

[2 marks]

[Total 10 Marks]

[END OF PAPER]